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Spatial determinants of U.S. FDI and exports in OECD countries

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ABSTRACT

World foreign direct investment (FDI) flows have been characterized by an unprecedented growth over the last four decades, with an uneven distribution across countries. This paper analyzes the spatial interdependence/third-country effects of multinational enterprises (MNE) for affiliate location choice and spillover effects of foreign direct investment (FDI). We specify a simultaneous equations model estimated using the generalized spatial two-stage least squares method on a sample of 25 OECD recipient countries of U.S. FDI during the period spanning the years 1999–2009. Our results lend support to the existence of complex vertical FDI and the significance of third-country effects, which is indicative of agglomeration economies.

1. Introduction

The last four decades have been characterized by an unprecedented growth of foreign direct investment (FDI). Data from the United Nations Conference on Trade and Development (UNCTAD) suggest that global FDI has increased at an average annual rate of about 14 percent. This outstanding performance in global FDI growth is greater than the growth of world GDP and global trade flows. As with most economic activities, the upward trend has not been consistently smooth. For example, Fig. 1 documents the trends of world inward and outward FDI flows over the last decade.

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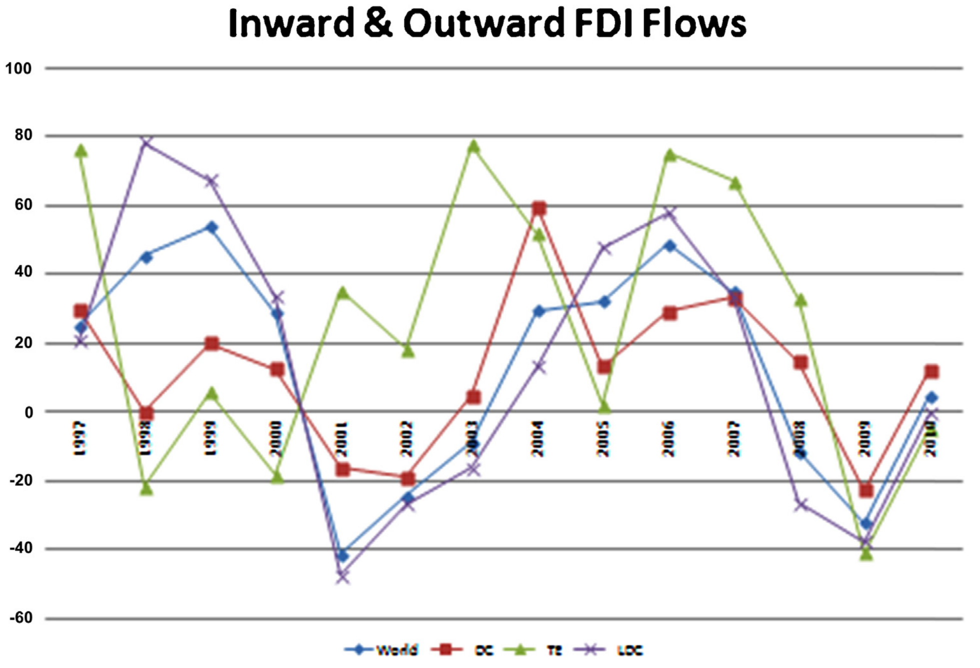

Fig. 1. Trends in FDI outflows: a regional comparison.

It appears that a historic peak was reached in 2007, followed by a sudden decline, which started in 2008 amidst turmoil in the global financial markets, and a weak recovery starting in late 2010. The background of a fragile world economy marked by fears of debt crisis in Europe and a hesitant U.S. economy might continue to weaken investors' confidence and dampen their enthusiasm for cross-country investments in developed economies. However, individual countries' experiences vary and so do the motives behind the decision to invest in a specific country (Walkenhorst, 2004). These cross-country variations in FDI patterns justify an analysis of outward FDI.

While FDI is a valuable source of foreign capital which helps to spur domestic investments and improve the competitiveness of domestic firms by providing advanced managerial skills (Balasubramanyam et al., 1996; Héricourt and Poncet, 2009; Sun and He, 2014), empirical evidence of the significance of the determinants of FDI has yet to be conclusively established. The most certain fact is that FDI is a cross-border investment of "lasting interest" undertaken by multinational corporations (MNCs) in an existing enterprise where the direct investor owns at least 10% of the voting power (OECD, 2010). Broadly speaking, previous efforts to explain the sprouting attention of MNCs in investing abroad hinge on market access and comparative advantage motives.[1] The former is known as the proximity concentration trade-off and refers to horizontal FDI, where an MNC production facility is designed to serve customers in a foreign market to avoid higher transportation costs and trade barriers (Brainard, 1997). The latter is fostered by the need to capitalize on differences in international factor prices by engaging in unskilled labor-intensive production in an unskilled labor-abundant country, and gives rise to what is known as vertical FDI (Baltagi et al., 2008). Thus, in this traditional two-country model of FDI, the impetus behind each of the two types of foreign investment decisions is nurtured by the characteristics of the home and recipient countries only.

[1] Interested readers can refer to Baltagi et al. (2007) for detailed discussions of the two types of FDI and recent efforts to extend the literature.

Nonetheless, numerous empirical studies provide mixed findings on the existence of these two types of FDI. More importantly, recent trends in the choice of FDI affiliates raise the question of what drives the rapid growth rate of FDI in some regions while other regions see their sources of foreign investment inflows drying up. Perhaps other forces need to be explored. A closer look at the recent FDI patterns shows that MNCs are following a different direct investment strategy with characteristics borrowed from both horizontal and vertical FDI. This hybrid type of FDI, known as complex FDI, cannot be explained by the bilateral FDI framework – the standard approach in most previous studies. Until recently, only a few limited attempts have argued that the costs of transportation, the factor intensity of production, and the costs of investing abroad, as well as policies pursued by the host countries' neighbors are characteristics of complex FDI (Yeaple, 2003). Moreover, the distinctive approach in complex FDI is to fragment production between the parent and the recipient countries to cater to the needs of consumers in either the host country or a "third market". These recent waves of studies on the determinants of FDI lend support to the existence of interdependencies across countries, a linkage not explained by the bilateral models. The aim of this paper is to fill that gap.

We make two contributions to the existing literature on FDI. First, we extend the generalized spatial two-stage least squares (GS2SLS) model developed by Kelejian and Ingmar (2004) to analyze the relationship between FDI and exports using a simultaneous equations model. According to the proximity – concentration trade-off hypothesis, FDI and exports are substitutes (Belderbos and Sleuwaegen, 1998; Gopinath et al., 1999). In contrast, the spillover effects of multinational corporations on the productivity of local firms in the host countries resulting from vertical FDI give rise to complementarity between FDI and exports (Pfaffermayer, 1994; Brainard, 1997; Co, 1997; Marchant et al., 1999; Clausing, 2000; Jensen, 2002; Wilson, 2006). In a second line of inquiry, which reflects our second contribution to the literature, we test the presence of third-country effects. The spatial autoregressive term in spatial FDI models along with the estimated coefficient of the market potential variable are used to assess the contemporaneous correlation between the recipient country and the adjacent potential markets. Blonigen et al. (2007) explain that the agglomeration externalities arising from the parent country into the host country would be captured by a positive spatial lagged coefficient on the FDI variable, whereas such agglomeration effects from the host into its surrounding countries would be evidenced by a positive coefficient on the market potential variable. The recent theoretical literature investigating these separate additional determinants of FDI finds evidence which lends support to the existence of agglomeration effects of FDI at the national level.[2] In other words, there are substantial positive spillover effects from a given country to its neighbors which have led to a greater consensus regarding third-country effects as an overlooked determinant of FDI. However, only few attempts have been made to measure the importance of third-country effects (Coughin and Segev, 2000; Blonigen et al., 2007; Baltagi et al., 2007; Garretsen and Peeters, 2009), mostly because spatial econometrics is still in the embryonic stages in the FDI literature even though spatial econometric models have been extensively used in cross-country studies to identify the presence of externalities or spillover effects as well as to generate unbiased and consistent estimates (Anselin, 1988).

As noted by Anselin (2009), ignoring spatial correlation in empirical regional studies could result in either omitted variable bias, which leads to biased and inconsistent or inefficient estimates. This could also explain why two-country empirical FDI studies yield mixed findings more often than not.[3] For example, while Ma et al. (2000) find trade and FDI to be substitutes in developed countries and complements in developing countries, Blonigen (2001) finds substantial evidence suggesting that FDI and trade can be both substitutes and complements even among developed countries. Our results provide supporting evidence for the existence of complex vertical FDI and the significance of third-country effects of U.S. FDI outflows into OECD countries.

[2] However, some authors (Villaverde and Maza, 2002) found regional disparities within countries.

[3] Along the same line of thought, Hanousek et al. (2011) reviewed a large body of literature pertaining to the effects of FDI on economies of countries transitioning from a command system toward a market system and concluded that the choice of the research design matters.

The remainder of the paper is structured as follows. Section 2 specifies the empirical model and describes the dataset. Section 3 presents and discusses the empirical findings. Section 4 concludes the paper.

2. Data and methodological framework

Against the background of the existing literature,[4] our first specification model is a modified gravity model with the traditional variables widely used in the FDI literature (Markusen and Venables, 2000; Brainard, 1997; Eaton and Tamura, 1994). Specifically, we relate U.S. outward FDI at time t to partner j's population, GDP, trade-costs, skilled-labor endowment, market potential, and membership in NAFTA, as well as the distance between the U.S. and the host country and a quadratic time trend.

The dependent variable, U.S. outward FDI, measures annual data on U.S. direct investment abroad into 25 OECD host countries[5] covering the period 1999–2009, which is taken from the U.S. Bureau of Economic Analysis (BEA). U.S. direct investment abroad is the value of U.S. outward direct investment stock measured at the historical cost basis expressed in millions of dollars of operations of parent companies and their foreign affiliates (BEA, 2011). The population and GDP variables, derived from the World Bank's World Development Indicators (WDI), measure the size of the domestic market in the host country. The measure of trade-costs is calculated as the ratio of openness (sum of exports and imports) to GDP, with trade data taken from the United Nations Commodity Trade Statistics Database (UNCOMTRADE). The host country's endowment in skilled workers, a proxy for human capital required for skilled-labor intensive production, is from the Barro and Lee dataset on educational attainment in the World from 1950 to 2010. It measures the average years of schooling for those who are 25 years of age and older, and is reported every five years. We use the linear interpolation method for the missing years. The measurement of market potential follows Head and Mayer (2004) and Blonigen et al. (2007).[6] The market potential variable for a country j is the sum of country j's GDP and the inverse-distance weighted GDP[7] for all other host countries in the sample, by year. Previous studies have found membership in regional trade blocs to significantly foster international trade relations among member countries. Following Baltagi et al. (2008), we use the variable NAFTA to account for membership in the same regional trade agreement, a dummy variable taking the value of 1 if the U.S. and the host country are both members of NAFTA and 0 otherwise. As in the traditional gravity model, the variable distance, measured between the U.S. and the host country, is used as a proxy for both higher management costs and higher trade-costs. The distance between the U.S. and the host country, measured in kilometers (km), is derived from the Centre d'Etudes Prospectives et d'Informations Internationales (CEPII). Table 1 provides the summary statistics of the variables included in our study.

Following Blonigen et al. (2007), we combine all the aforementioned variables and specify the following spatial autoregressive model where all the variables are in natural logs except for the time trend and the dummy variables:

$$\begin{aligned}\mathrm{FDI}_{ijt} = {} & \alpha_0 + \rho W_y \mathrm{FDI}_{ijt} + \alpha_1 \mathrm{GDP}_{jt} + \alpha_2 \mathrm{Distance}_{ij} + \alpha_3 \mathrm{Population}_{jt} + \alpha_4 \text{host skilled}_{jt} \\ & + \alpha_5 \text{Trade cost}_{jt} + \alpha_6 \text{mrkt potential}_{jt} + \alpha_7 \mathrm{NAFTA}_{ij} + \alpha_8 \mathrm{Time} + \alpha_9 \text{Time squared} \\ & + \varepsilon_{it} \end{aligned} \tag{1}$$

where the subscript i stands for the U.S., $j = 1, 2, \ldots, 25$, represents the host countries and $t = 1, 2, \ldots, T$ the years 1999–2009; the other variables are as already defined.

[4] See Faeth (2009) for an extensive literature review on the relevant determinants of FDI derived from various theoretical models. He concludes that "there is no single theory of FDI, but a variety of theoretical models attempting to explain FDI and the location decision of MNEs."

[5] Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Japan, Korea, rep., Luxembourg, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Spain, Sweden, Switzerland, Turkey, and United Kingdom.

[6] They find that regions with high GDP and/or regions surrounded by larger markets attract more FDI.

[7] We use the same set of weights for the construction of inverse-distance weighted GDP as for the construction of the spatial lag term, which we will discuss later.

Table 1
Descriptive statistics of the variables from 1999 to 2009.

Variable	Mean	Std. deviation	Minimum	Maximum
FDI (in millions)	60,384.11	85,157.61	760	449,521
Trade (in thousands)	2.23e+07	3.88e+07	274,241.2	2.22e+08
GDP (in millions)	685,205	1,015,066	18,691	5,201,164
POP (in millions)	34,101.63	34,430.4	430	127,773
Distance (in km)	6407.889	2128.246	548.39	14,546.24
MP (in millions)	1,894,279	1,234,849	165,960.4	5,942,495
Language	0.13	0.34	0	1
Colony	0.12	0.32	0	1
NAFTA	0.83	0.27	0	1

Note: MP is market potential.

Eq. (1) follows a spatial autoregressive model. Various spatial weight matrices have been used in the spatial econometric literature, including the binary contiguity matrix (Anselin and Arribas-Bel, 2013; Birkelof, 2010) or more generalized distance-based weight matrices (Blonigen et al., 2007; Conley and Ligon, 2002; Gallo and Kamarianakis, 2011). Unlike the contiguity matrix, the distance weight matrix guarantees connections between countries located in different continents such as Japan and European countries or South Korea and Canada, for example. In this paper, we consider a weight matrix, W_y, which clusters the geographical distance between any pair of host countries. The elements of W_y, $W_y(d_{j,k})$, are constructed by dividing the shortest distance of the sample by $(d_{j,k})$ – the distance between a pair of host countries. The shortest distance in our sample is 173.033 km, separating Brussels and Amsterdam. Specifically,

$$W_y(d_{j,k}) = \frac{173.033}{d_{j,k}} \quad \forall j \neq k \tag{2}$$

where $W_y(d_{j,j}) = 0$, and $W_y(d_{j,k}) = 1$ when $(d_{j,k})$= the distance between Brussels and Amsterdam, and all other host country pairs receive a weight that declines with distance. Thus, our weight matrix is:

$$W_y = \begin{pmatrix} 0 & W_y(d_{i,j}) & W_y(d_{i,k}) \\ W_y(d_{j,i}) & 0 & W_y(d_{j,k}) \\ W_y(d_{k,j}) & W_y(d_{k,j}) & 0 \end{pmatrix} \tag{3}$$

The standard approach in spatial econometrics is to row normalize the weight matrix so that each row sums to unity.

The estimation of Eq. (1) forms our baseline results. While previous studies have succeeded in establishing FDI interdependence amongst potential host countries, they rely on a strong assumption on the causation between FDI and exports. Thus, to avoid making any assumptions on the correlation between FDI and exports, we specify a simultaneous equation model of FDI and exports. This framework allows for greater flexibility, leaving room for plausible feedback between FDI and exports. The spatial simultaneous equations model, our preferred model, is expressed as follows:

$$\begin{aligned}\text{Exports}_{ijt} = {} & \alpha_{11} + \rho_1(I \otimes W_y)\text{Exports}_{ijt} + \alpha_{12}\text{FDI}_{ijt} + \alpha_{13}\text{GDP}_{jt} + \alpha_{14}\text{Distance}_{ij} \\ & + \alpha_{15}\text{language}_{ij} + \alpha_{16}\text{Colony}_{ij} + \mu_{1it}\end{aligned} \tag{4}$$

$$\begin{aligned}\text{FDI}_{ijt} = {} & \alpha_{21} + \rho_2(I \otimes W_y)\text{FDI}_{ijt} + \alpha_{22}\text{Exports}_{ijt} + \alpha_{23}\text{GDP}_{jt} + \alpha_{24}\text{Population}_{jt} \\ & + \alpha_{25}\text{Market potential}_{jt} + \alpha_{26}\text{Distance}_{ij} + \alpha_{27}\text{NAFTA}_{ij} + \mu_{2it}\end{aligned} \tag{5}$$

where the variables FDI, GDP, distance, population, market potential and NAFTA are defined as in Eq. (1); language is a dummy variable which equals 1 when English is the official language in the host country and 0 otherwise; colony is a binary variable that takes the value of 1 if the U.S. and the host country share a colonial tie and 0 otherwise; ε is an error term; W_y, an inverse distance weight matrix of the dimension $n \times n$, is the same in all three equations; ρ is the spatial autoregressive coefficient to be estimated, and its value is assumed to lie between -1 and 1; μ_1 and μ_2 are error terms. I is the identity matrix of dimension T, and $\otimes$ is the Kronecker product. Since the W_y is row standardized, then W_yFDI and $(I \otimes W_y)$FDI are interpreted as row-sums being a proximity-weighted average of FDI into alternative countries (Blonigen et al., 2007), and $(I \otimes W_y)$ exports is the weighted average of neighboring countries' exports (Porojan, 2001).

The system of Eqs. (4) and (5) makes FDI and exports endogenous variables, and its estimation requires the use of instrumental variables. However, Kelejian and Ingmar (2004) point out that the inverse distance weight matrix defined above and the exogenous variables in each equation represent an instrument matrix for estimation purposes. Moreover, in a linear simultaneous equations model, Greene (2003) states that the order condition, which is a necessary but not a sufficient condition, requires that the number of exogenous variables excluded from one equation must be at least as large as the number of dependent variables included in that equation. Thus, the order condition of the above defined system of equations is fulfilled because Eq. (4) excludes two variables, while Eq. (5) excludes three variables.

Finally, the simultaneous equations model is estimated by generalized spatial two-stage least squares using a three-step procedure. In the first step, the equations are estimated by two-stage least squares (2SLS) using a matrix of exogenous variables and spatially lagged exogenous variables as an instrument. In the second step, the autoregressive parameter ρ is estimated by the generalized method of moments (GMM) procedure following Kelejian and Ingmar (1999). In the third step, we use the Cochran–Orcutt transformation to estimate ρ. The estimated value of ρ indicates the existence of spatial autocorrelation in the dataset.

3. Empirical results and discussion

Our benchmark results are the estimations of two variants of Eq. (1). The first specification shows the coefficients for the standard set of variables included in the gravity model of FDI augmented with the market potential variable, time trend and time trend squared to allow a comparison with the findings by Blonigen et al. (2007). The second specification adds the variable NAFTA to the analysis, which is a way to account for the reduction in trade barriers amongst countries belonging to the same regional trade bock (Buch and Piazolo, 2001). The empirical results are presented in Table 2.

Both specifications show that the coefficients on the traditional determinants of FDI are statistically significant at the 1% level, with the exception of the host country's endowment in skilled workers. More importantly, the effects of the variables are in line with the literature on FDI. The main results of interest in Table 2 are the estimated parameters related to the market potential and the spatial lagged FDI. The positive and statistically significant coefficients of the spatially lagged variables suggest the presence of complex FDI, which is consistent with Blonigen et al. (2007). Specifically, the spatial lag has a point estimate of 0.43 in the first specification and 0.37 in the second specification. For example, the first specification suggests that a 10 percent increase in FDI in other proximity-weighted countries increases FDI in a host country by 4.3 percent, everything else being equal. The positive value of the spatial lag lends support to the existence of vertical FDI. In contrast, we obtain startling results with respect to the coefficient of market potential in both specifications. Not only is the estimated parameter surprisingly negative, but it is also not robust to the inclusion of other variables such as NAFTA. Specification 1 shows that market potential is statistically negative with an elasticity of -0.17. In other words, a 10 percent increase in the market potential around any given host country decreases FDI in that host country by 1.7 percent, ceteris paribus In contrast, specification (2) reveals that market potential comes up as statistically not significant. Two lines of thought could help explain the results of the second specification. One observation is that NAFTA is picking up the effects of market potential, as the goal of NAFTA as a trade bloc is to eliminate barriers to trade and investment amongst member countries (which happen to be in the same geographic area). Another observation, as noted by Blonigen et al. (2007), is that the inclusion of the host GDP in computing the market potential variable may overstate the estimated coefficient of the variable. However, the exclusion of the host country's GDP from the market potential variable fails to improve the statistical significance of the variable. While the coefficients on other variables remain stable, the coefficient of the NAFTA variable turns insignificant. The results are available upon request.

Table 2
Spatial autoregressive model using data from 1999 to 2009.

Variables	(1)	(2)
GDP	1.54***	1.54***
	(12.66)	(12.74)
Host country skill	0.01	−0.08
	(0.02)	(0.22)
Trade-cost	−0.80***	−0.71***
	(−3.82)	(−3.15)
Population	−0.81***	−0.86***
	(−7.51)	(−7.48)
Distance	−0.82***	−0.70***
	(−8.09)	(−5.22)
Market Potential	−0.17*	−0.14
	(−1.75)	(−1.33)
NAFTA	–	0.62**
		(2.02)
Time	0.03	0.02
	(0.29)	(0.25)
Time 2	−0.001	−0.00008
	(−0.16)	(−0.07)
ρ	0.43***	0.37***
	(4.24)	(3.38)

Note: *T* statistics are in parentheses.
* Significance at the 10% level.
** Significance at the 5% level.
*** Significance at the 1% level.

An important and general observation is that the empirical study of FDI suffers from specification issues which could potentially invalidate the importance of many determinants of FDI. Moreover, the relationship between FDI and exports remains unsettled. For example, Egger and Pfaffermayr (2004) use a seemingly unrelated regression model applied to bilateral industry level exports and outward stocks of FDI from the U.S. and Germany to other OECD and non OECD countries between 1989 and 1999 to investigate the relationship between distance, trade and FDI. They conclude that exports and FDI may be substitutes or complements with respect to distance, depending on the relative factor endowments. Also, Blonigen and Tomlin (2001) find substantial evidence of the presence of both substitute and complementary relationships between trade and FDI based on industry-level FDI inflow from Japan to the U.S. between 1980 and 1990. The case for spatial considerations has been made by such authors as Blonigen et al. (2007), Baltagi et al. (2008), and Garretsen and Peeters (2009). However, a mostly ignored issue is the possibility of feedback effects between FDI and exports. A simultaneous equation model is the most attractive specification to address this issue. To that end, we fit the model represented by Eqs. (4) and (5). The results are summarized in Table 3.

Column 1 displays the results of the FDI equation, Eq. (5). Eq. (5) modifies Eq. (1) by excluding the quadratic time trend and substituting trade-costs with exports. A comparison between the estimated results from Eqs. (1) and (5) yields two important observations. First, the spatial lag parameter and the coefficient on the market potential variable are both statistically positive and economically significant. This suggests a vertical specialization with agglomeration externalities. The results are in line with Garretsen and Peeters (2009), who support this finding for Dutch FDI into OECD countries. However, our findings are in contrast with those of Blonigen et al. (2007), who found export platform FDI for OECD countries. The relationship between FDI and exports is highlighted in the system of equations with positive and statistically significant coefficients, as shown in both columns 1 and 2 in Table 3. For instance, column 1 indicates that a 10% increase in exports causes FDI to increase by 5.9%. Moreover, column 2, which presents the result for the exports equation, reinforces the result from column 1. The parameter estimate suggests that a 10% increase in FDI causes a 2.5% increase in exports. This positive feedback lends support to a complementary relationship between FDI and exports. These results corroborate the findings by Clausing (2000), who concludes that U.S. FDI and exports to OECD countries are complementary.

Table 3
Generalized spatial two-stage least squares using data from 1999 to 2009.

Variables	FDI (1)	Exports (2)
Constant	−9.24***	−3.91**
	(−3.78)	(−2.37)
GDP	0.67***	0.89***
	(4.72)	(25.22)
Population	−0.66***	–
	(−7.11)	
Distance	−0.38**	−0.64***
	(−2.09)	(−8.22)
Language	–	0.32***
		(2.52)
Market potential	0.20**	–
	(2.26)	
Colony	–	−0.74***
		(−6.09)
NAFTA	−0.17	–
	(−0.35)	
Exports	0.59***	–
	(6.41)	
FDI	–	0.25***
		(8.10)
ρ	0.81***	0.72***
	(6.88)	(7.00)

Note: *T* statistics are in parentheses.
* Significance at the 10% level.
** Significance at the 5% level.
*** Significance at the 1% level.

Table 4
Generalized spatial two-stage least squares using data from 1999 to 2007.

Variables	FDI	Export
Constant	−8.51***	−5.22***
	(−3.36)	(−2.64)
GDP	0.75***	0.91***
	(4.57)	(22.69)
Population	−0.64***	–
	(−6.19)	
Distance	−0.38**	−0.53***
	(−2.09)	(−6.97)
Market potential	−0.02	–
	(−0.24)	
Language	–	0.62***
		(5.23)
Colony	–	−0.76***
		(−5.68)
NAFTA	−0.51	–
	(−1.01)	
Export	0.54***	–
	(5.77)	
FDI	–	0.21***
		(6.02)
ρ	1.04***	0.77***
	(8.19)	(6.02)

Note: *T* statistics are in parentheses.
* Significance at the 10% level.
** Significance at the 5% level.
*** Significance at the 1% level.

To isolate the potential effects of the 2008 financial crisis on U.S. outward FDI, we re-estimated our simultaneous equations model using data spanning the years 1999–2007. The results are summarized in Table 4.

Two important observations can be made from Table 4. The estimated coefficients on the traditional variables in the FDI and exports equations remain statistically and economically robust. In contrast, the market potential variable is no longer statistically significant, which lends support to the existence of pure vertical FDI rather than complex vertical FDI. This unexpected result raises the question as to whether the significance of market potential depends on the sample and time frame under study. Further research could help shed light on this important issue.

4. Conclusion

Most previous studies use a two-country framework to analyze the determinants of FDI and the relationship between FDI and exports in a single equation model. For the most part, the results have been inconclusive. Recent developments in the field of trade and financial flows point to the existence of third-country effects as an overlooked determinant of FDI. In this paper, we reexamine the single equation spatial autoregressive FDI model following Blonigen et al. (2007) as our benchmark results. While the results support the existence of third-country effects, we find that the estimated coefficient of the market potential variable has an unexpected negative sign. In all likelihood, this startling result could be due to the endogeneity of the trade variable in the FDI equation. To circumvent the issue, we specify a simultaneous equations model of FDI and exports within a spatial econometric framework. This specification allows for greater flexibility by leaving room for possible feedback between FDI and exports. The results of the simultaneous equations model confirm the significance of third-country effects. In addition, our results strongly underpin the existence of agglomeration effects, as evidenced by the statistically significant positive coefficients on the market potential and spatial lag variables. The simultaneous equations model of FDI and exports develops a tractable way of describing complex vertical FDI from the U.S. into other OECD countries. Moreover, we find evidence which is suggestive of a complementary relationship between FDI and exports. These findings suggest that host countries and adjacent locations will benefit from attracting more FDI because of the spillover effects which help to improve productivity and spur economic growth.

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